2024
NVIDIA Corporation
Annual Review

Notice of Annual Meeting
Proxy Statement
Form 10-K

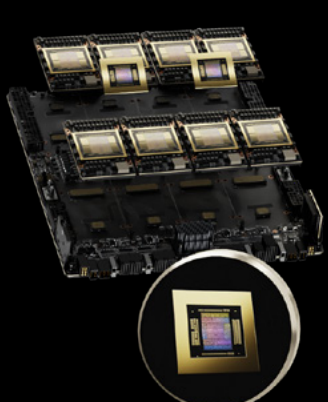
“The sum of all that NVIDIA’s doing will indeed create the next industrial revolution”

CNBC

Accelerated computing is sustainable computing. Every data center in the world needs to be accelerated to reclaim power, achieve sustainability, and realize net-zero emissions. Accelerated data centers could save an incredible 19 terawatt-hours of electricity annually if run on GPU and DPU accelerators vs CPUs. That’s about the same energy as a year’s worth of trips by 2.9 million passenger cars.

The efficiency of accelerated computing paved the way for generative AI. The most critical computing platform of our generation, generative AI will reshape the world’s largest industries and create an entirely new one.

NVIDIA, the pioneer of accelerated computing, is the driving force of this new era.



HGX B100

NVLINK Switch

GB200 Superchip
Compute Node

“They basically have a comprehensive solution from the chip all the way to data centers at this point”

CIO

Accelerated computing starts with the most advanced processors and ends with AI factories. From chip architecture to advanced networking to acceleration libraries, NVIDIA builds the entire computing system at data-center scale. Then, we disaggregate everything and reintegrate it into the world’s computing fabric so that industries can leverage the parts and systems they need.

In the future, almost all of our experiences will be generative. Blackwell—the world’s most powerful AI platform—is tailor-made for the generative AI revolution.

Quantum X800 Switch
ConnectX-8 SuperNIC

Spectrum X800 Switch
BlueField-3 SuperNIC

"Continually optimized software remains NVIDIA's ace in the hole"

Forbes

Accelerated computing requires full-stack software. NVIDIA's acceleration stacks optimize workloads on a massive scale, integrating thousands of nodes while treating network and storage as integral components.

This year, we rolled out TensorRT-LLM and NVIDIA Inference Microservices™ (NIM). TensorRT-LLM is an open-source software library that enables customers to more than double the inference performance of their GPUs. NIM are a new way to package and deliver AI software. This curated selection of microservices adds a new layer to NVIDIA's full-stack computing platform—connecting the AI ecosystem of model developers, platform providers, and enterprises with a standardized path to run custom AI models.

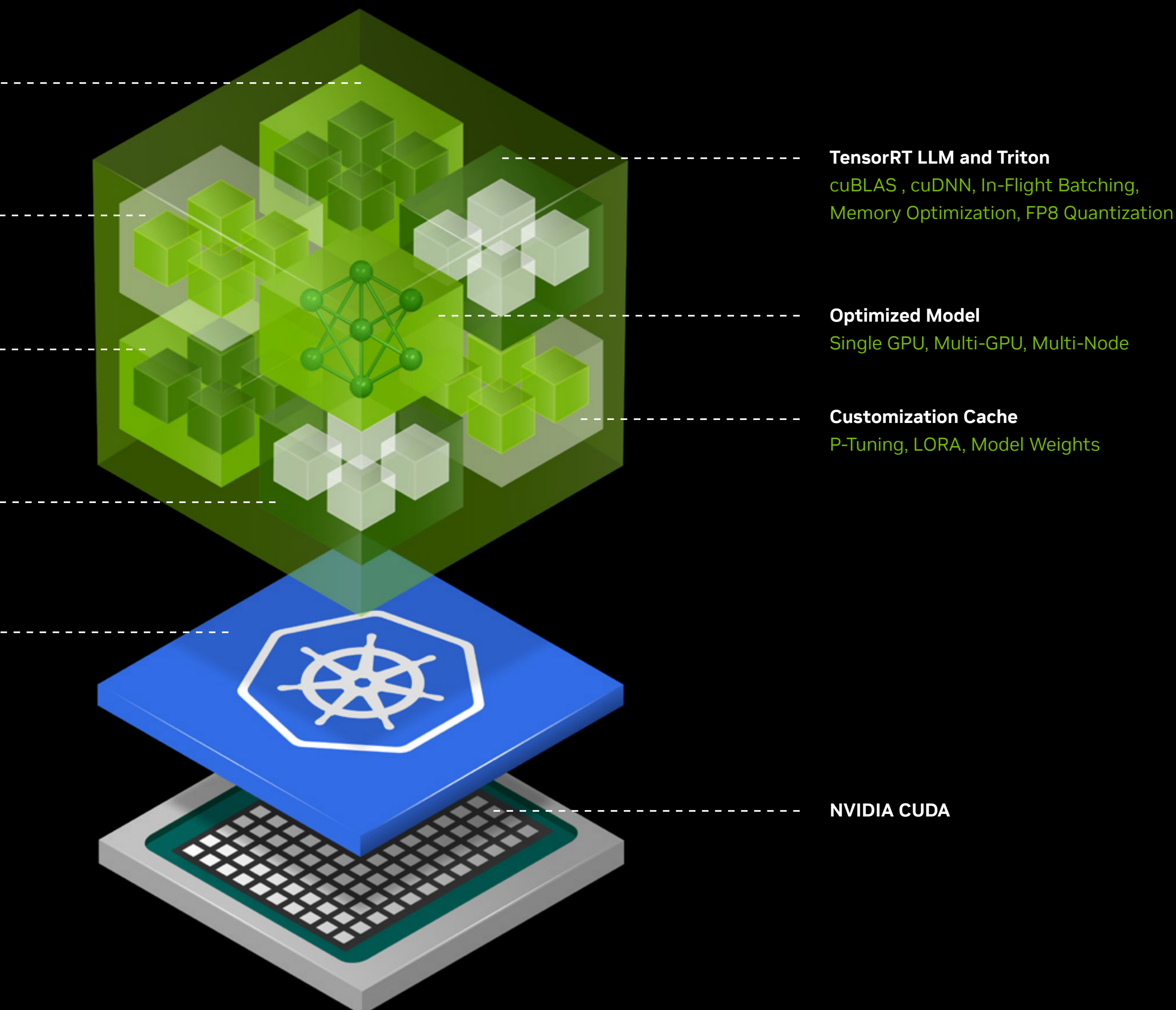
Industry Standard APIs
Text, Speech, Image, Video, 3D, Biology

Triton Inference Server
cuDF, CV-CUDA, DALI, NCCL, Post Processing Decoder

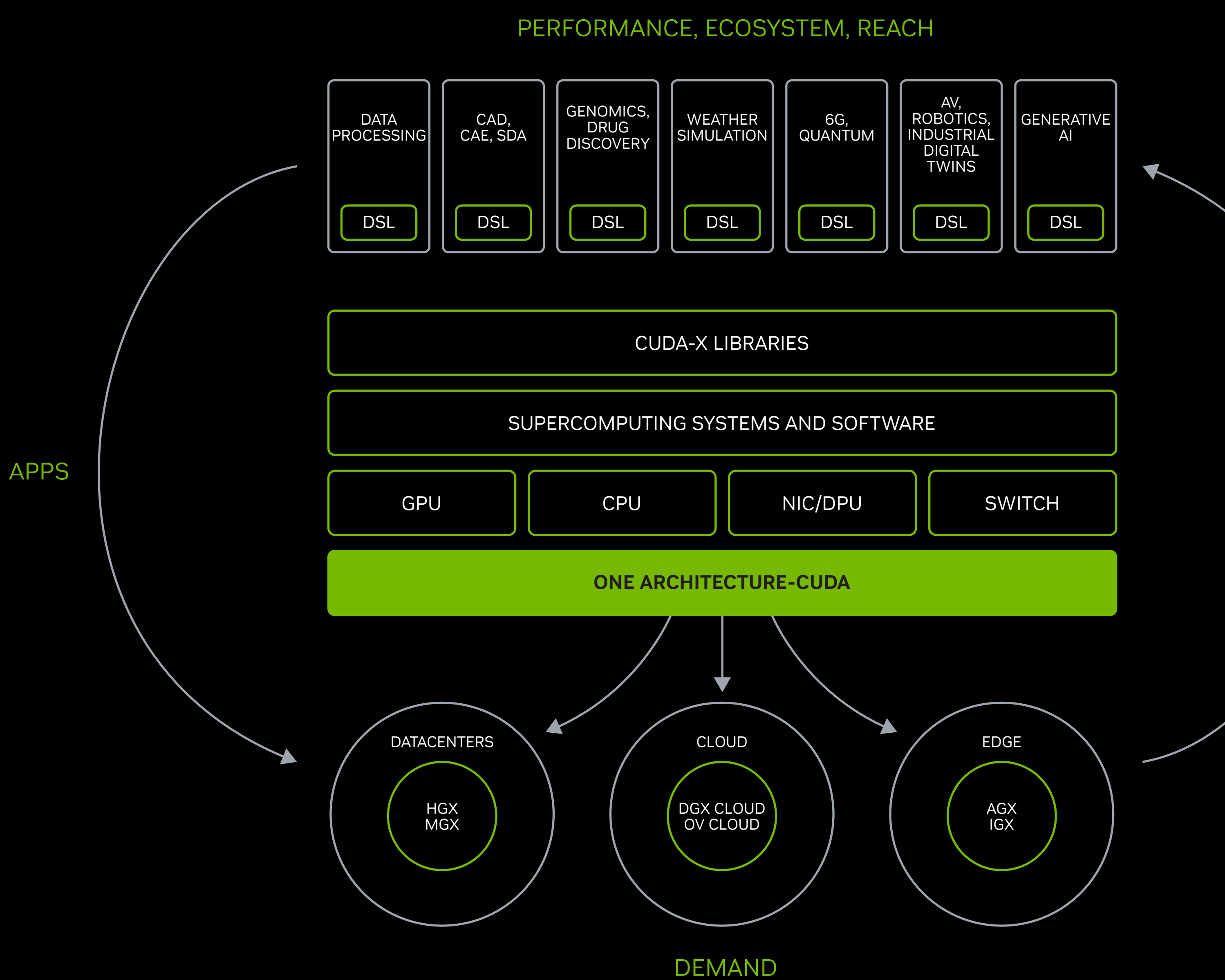
Cloud Native Stack
GPU Operator, Network Operator

Enterprise Management
GPU Health Check, Identity, Metrics, Monitoring, Secrets Management

Kubernetes



100’s of Millions of CUDA GPUs Installed Base


PERFORMANCE, ECOSYSTEM, REACH
DATA PROCESSING
DSL
CAD, CAE, SDA
DSL
GENOMICS, DRUG DISCOVERY
DSL
WEATHER SIMULATION
DSL
6G, QUANTUM
DSL
AV, ROBOTICS, INDUSTRIAL DIGITAL TWINS
DSL
GENERATIVE AI
DSL
APPS
CUDA-X LIBRARIES
SUPERCOMPUTING SYSTEMS AND SOFTWARE
GPU
CPU
NIC/DPU
SWITCH
ONE ARCHITECTURE-CUDA
DATACENTERS
HGX
MGX
CLOUD
DGX CLOUD
OV CLOUD
EDGE
AGX
IGX
DEMAND

INSTALLED BASE

“NVIDIA’s got great chips, and more importantly, they have an incredible ecosystem”

The New York Times

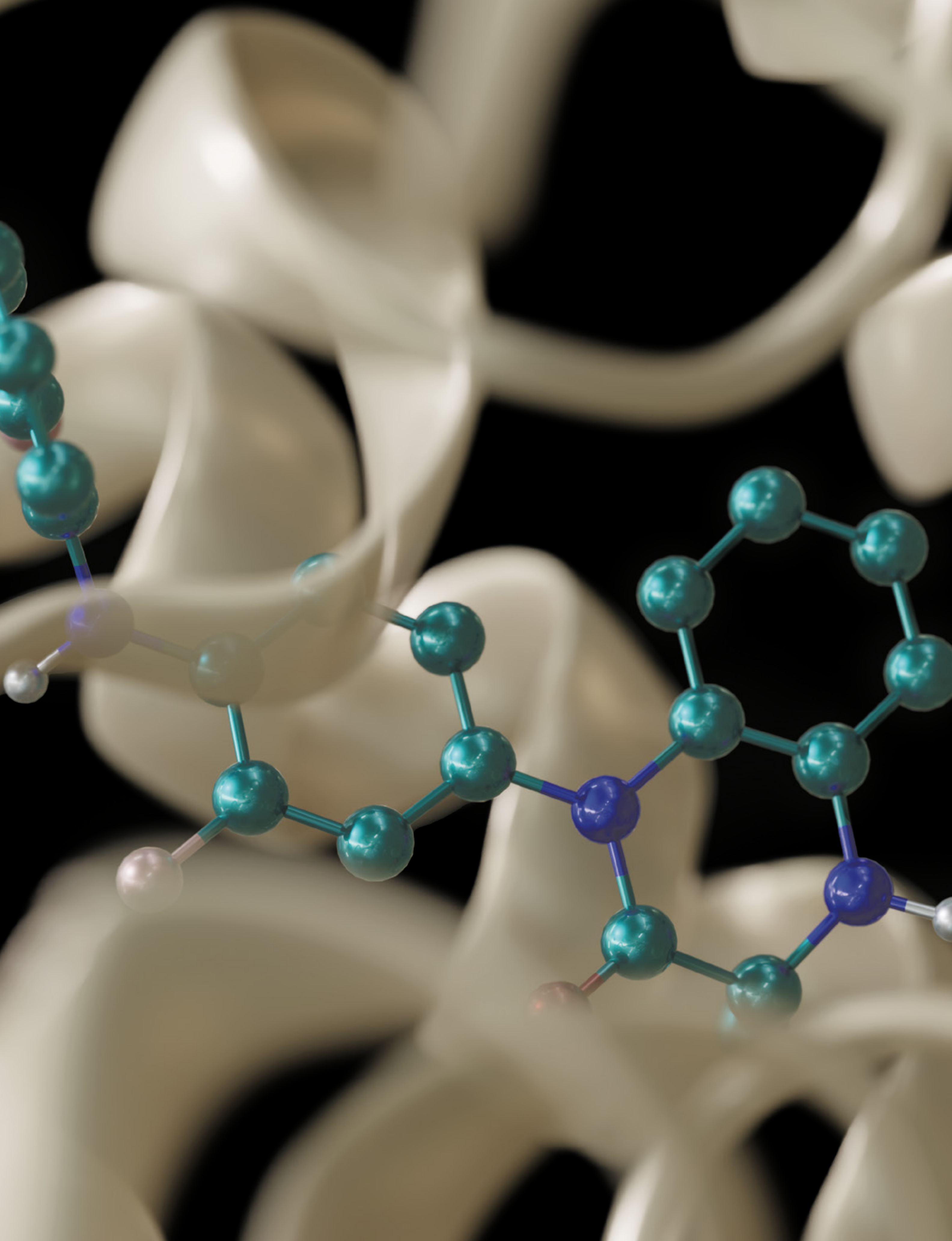
NVIDIA’s accelerated computing ecosystem is bringing AI to every enterprise. The NVIDIA ecosystem spans nearly **5 million** developers and **40,000** companies. More than **1,600** generative AI companies are building on NVIDIA. CUDA®, our parallel computing model launched in 2006, offers developers more than **300** libraries, **600** AI models, numerous SDKs, and **3,500** GPU-accelerated applications. CUDA has more than **48 million** downloads.

“NVIDIA’s prescription for the future: transforming healthcare with AI”

Forbes

NVIDIA AI is powering the next era of drug discovery and advances in life sciences. NVIDIA Clara™, our suite of computing platforms, software, and services for healthcare and life sciences, and NVIDIA BioNeMo™, our platform for state-of-the-art generative AI models for drug discovery, are turbocharging breakthroughs.

Genentech is tapping NVIDIA to use generative AI to discover and develop new therapeutics and deliver treatments to patients more efficiently. Recursion Pharmaceuticals is the first NVIDIA partner to offer an AI model through BioNeMo cloud APIs. And Amgen is building AI models trained to analyze one of the world’s most extensive human datasets on an NVIDIA DGX SuperPOD™.



> “NVIDIA has virtually recreated the entire planet—and now it wants to use its digital twin to crack weather forecasting for good”
>
> ***TechRadar***

NVIDIA AI is tackling climate change. Extreme events attributable to climate change cost more than $143 billion a year. NVIDIA’s CorrDiff is a revolutionary new generative AI model trained on high-resolution radar, weather forecasts, and other data. Using CorrDiff, extreme weather events can be super-resolved from 25-kilometer to two-kilometer resolution with 1,000 times the speed and 3,000 times the energy efficiency of conventional weather models. This AI-powered weather forecasting allows us to more accurately predict and track severe storms to try and reduce those impacts.

“NVIDIA is a core technology leader in the autonomous driving arena”

CleanTechnica

NVIDIA AI is revolutionizing transportation. NVIDIA has developed an end-to-end platform from cloud to car to allow the automotive ecosystem to develop industry-leading AI-defined vehicles. We recently announced DRIVE AGX Thor™—designed and optimized for gen AI utilizing NVIDIA’s Blackwell architecture—to reimagine the driving experience.

And this year, every next-generation Mercedes-Benz vehicle will include an NVIDIA software-defined computing architecture that includes the most powerful computer, system software, and applications for consumers.


“The age of humanoid robots could be a significant step closer thanks to a new release from NVIDIA”

TechRadar

NVIDIA is fueling the next wave of AI—robotics and industrial digitalization. And that new wave of robots that will learn in NVIDIA Omniverse. Simulators like Isaac Sim running on Omniverse will be gyms where robots learn their skills. Over 1.2 million developers and 10,000 customers and partners are leveraging the NVIDIA Isaac and Jetson platforms to create and deploy AI-driven robots. And Project GR00T, a general-purpose foundation model for humanoid robots, will help them understand natural language and emulate movements by observing human actions.


A1

“NVIDIA Omniverse Cloud APIs will elevate digital twins for a new industrial revolution”

Venture Beat

The soul of NVIDIA is where computer graphics, physics, and AI intersect in Omniverse—a virtual world simulation engine. Heavy industry is one of the final frontiers of IT. Omniverse is the fundamental operating system for building digital twins that are crucial to unlocking new potential in heavy industries worldwide.

Omniverse connects the tools created by developers in the NVIDIA ecosystem. It enables each team to operate on the same ground truth, creating efficiencies and innovation. And now, with NVIDIA Omniverse Cloud™ APIs, developers can simplify and speed up the development of digital twins for almost any industrial application, seamlessly integrating Omniverse into their existing apps.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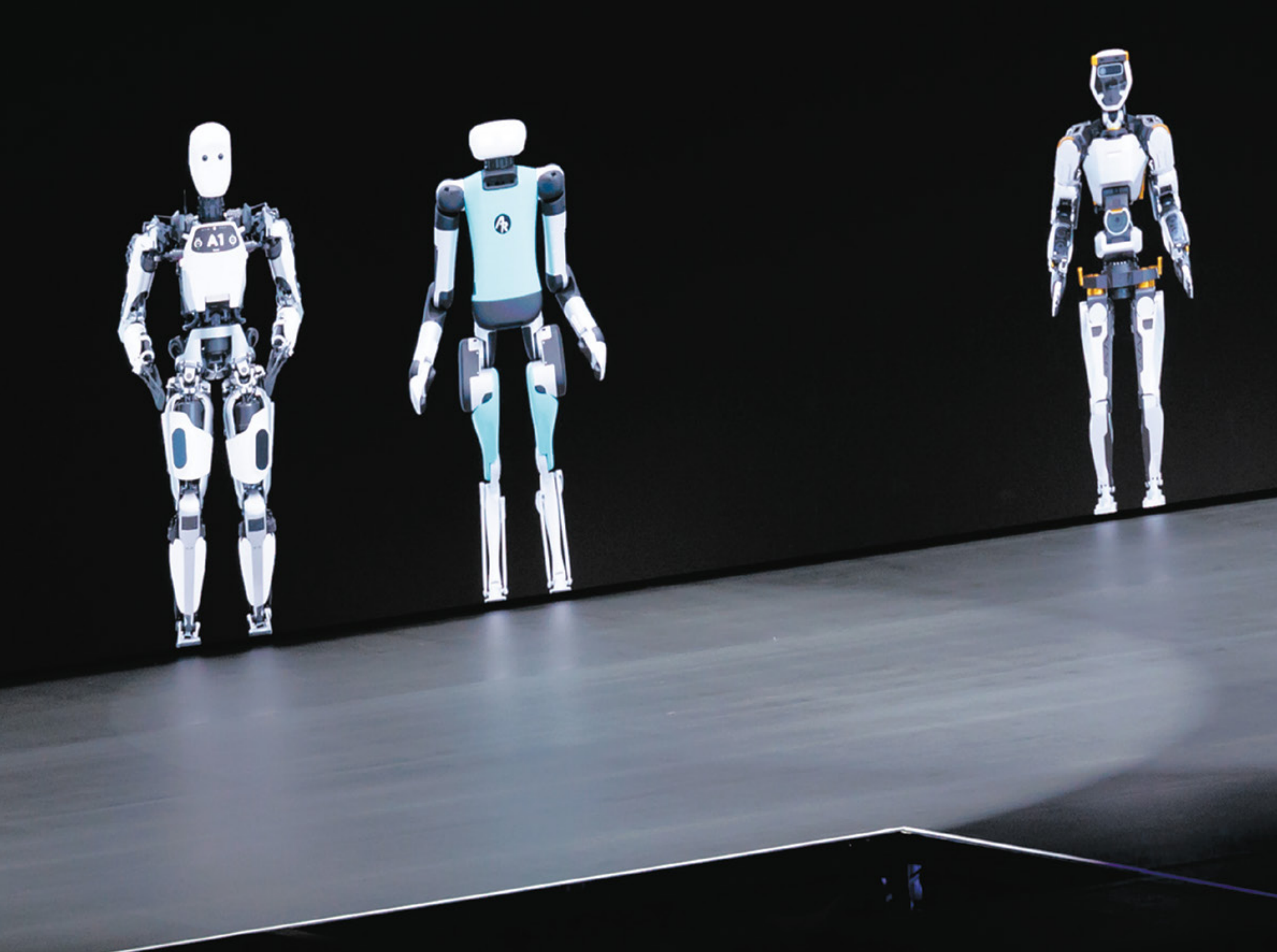
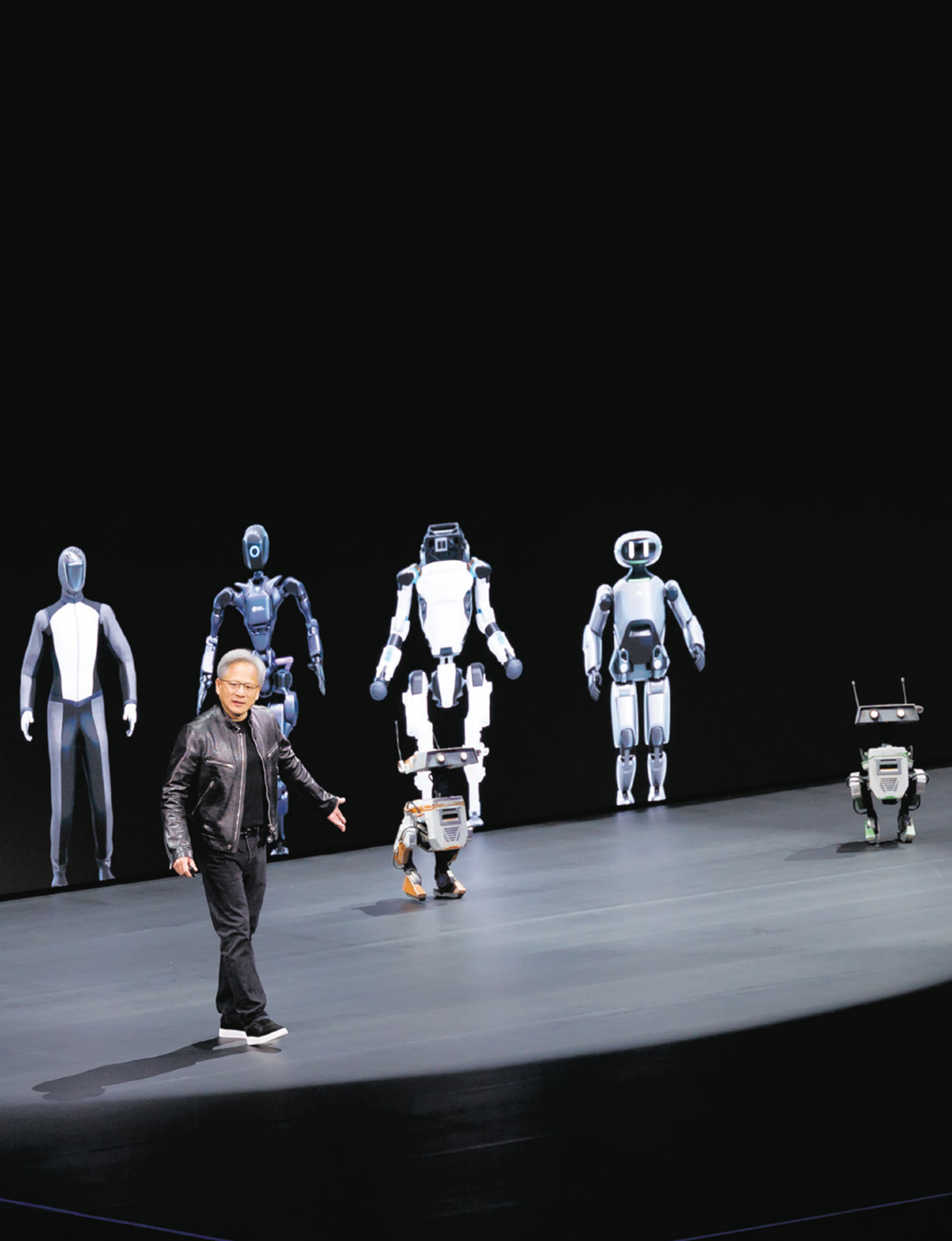
"Generative AI is the defining technology of our time. Blackwell is the engine to power this new industrial revolution. Working with the most dynamic companies in the world, we will realize the promise of AI for every industry."





Dear NVIDIANs and Stakeholders,

The dawn of a new computing era is unmistakably upon us.

Two transitions are occurring simultaneously—accelerated computing and generative AI—transforming the computer industry and every other industry worldwide.

CPU scaling, the engine that has driven the computer industry over the past three decades, has slowed. Software developers have turned to accelerated computing to keep pace with the exponentially growing demand for computing in a cost- and energy-sustainable way. The computing model pioneered by NVIDIA has reached critical mass and has been adopted broadly as the path forward.

Accelerated computing will modernize the world's trillion-dollar data center infrastructure. Its increased throughput and cost and energy savings are remarkable. Two NVIDIA HGX computers with Hopper GPUs that together cost $500K can replace 1,000 nodes of CPU servers that cost $10M for AI, scientific computing, or data processing workloads.

For data centers that are budget- or power-limited, NVIDIA delivers an order-of-magnitude increase in throughput. For data centers optimized for throughput, NVIDIA delivers an order-of-magnitude reduction in cost and power. Accelerated computing is the best way to build sustainable data centers.

When the cost of a fundamental resource, like computing, improves by orders of magnitude, new methods are invented, and new utilities are discovered. AI researchers tapped NVIDIA CUDA to realize deep learning, a machine learning algorithm that is incredibly compute-intensive. Deep learning processes mountains of data to find patterns and relationships and learn predictive features. In 2012, AlexNet shocked the artificial intelligence community by winning, by a considerable margin, the ImageNet computer vision contest. Within a couple of years, every computer vision algorithm used deep learning, and within five years, computer vision had achieved superhuman object recognition capabilities. AlexNet on NVIDIA CUDA was the big bang of modern AI.

We recognized that AlexNet was more than a computer vision breakthrough. We imagined deep learning would fundamentally revolutionize software and, consequently, all computing. Deep learning can scale to tackle many previously impossible challenges with new network architecture innovations, more data, and more compute.

In the ensuing decade, NVIDIA set out to advance AI. We pursued several strategies:

Reinvent every layer of computing, from GPU architecture, e.g., Tensor Cores, to system interconnect, e.g., NVLink, systems, e.g., DGX, software, e.g., cuDNN and TensorRT, to networking technologies, e.g., SHARP in-networking processing, and much more. We reinvented computing from the ground up.

Transform into a data-center-scale company by combining with Mellanox, the world's leading high-performance networking company. As computing large models and training data scales to require millions of trillions more operations, we need not just amazing chips but AI supercomputing systems. Today, NVIDIA is an expert at every computing layer, from chips to software, and across the data center—compute to networking. We have the expertise and scale to help customers build AI infrastructures anywhere.

Become an AI expert by building NVIDIA AI Research and application-specific AI platforms—full-stack AI platforms for DLSS RTX neural graphics, DRIVE AV self-driving cars, Isaac robotics, ACE digital humans, NeMo LLMs, BioNeMo digital biology, and Clara medical imaging—so that we can apply AI firsthand to create new markets. Automotive has joined gaming as a multi-billion-dollar run rate business. Healthcare and robotics are additional multi-billion-dollar opportunities.

Build a new ecosystem for the AI era by creating tools, libraries, and expert technical teams to support developers going after the AI revolution. The global NVIDIA ecosystem now approaches 5 million developers—capping a record-setting year for new developers. Forty thousand companies have worked with NVIDIA. There are now more than 3,300 GPU-accelerated applications. More than 18,000 startups are building on NVIDIA.

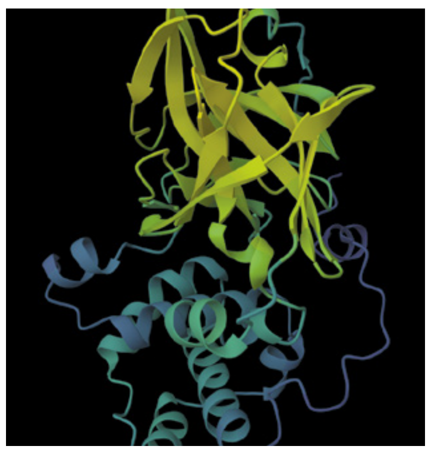
These four strategies—reinventing the full computing stack, building data-center-scale AI supercomputers, conducting basic AI research and applying AI to create new markets, and building a global ecosystem—transformed NVIDIA and all of computing.

Twelve years after AlexNet, NVIDIA is a full-stack, data-center-scale AI company that partners with nearly every AI company in every industry worldwide. Then ChatGPT arrived, and the pace of AI accelerated even faster. The era of generative AI began.

Computers can now generate information based on human prompts. At the heart of ChatGPT, large language models, or LLMs, learned to understand human language, prior knowledge, and even some common sense from massive amounts of text.

And not only human language. LLMs have learned to understand text, speech, images, and even amino acids and proteins, the language of biology. With multi-modal learning—for example, simultaneously learning from language and images—generative AI is gaining a deeper understanding of the world and can translate across different modalities.

Given a prompt such as "**golden retriever wearing sunglasses and a white golden doodle with black leather jacket surfing a wave, on a surfboard with NVIDIA logo, in Maui during sunset, with hotels on the beach**," generative AI can create an image.



From an amino acid sequence, AI can generate the structure of a protein, e.g., the SARS-CoV-2 main protease.

The combinations and possibilities are endless.

Generative AI is the most transformative and consequential technology of our generation. Computers will understand people and the world and perform tasks that previously required human intelligence. With generative AI, computers can augment and amplify humans to perform tasks with greater ease and speed, help us be more productive, and solve problems previously unimaginable.

Researchers use generative AI to compose billions of chemical compounds and virtually test them as drugs against a disease target. Climate scientists are using generative AI to map tens of thousands of potential trajectories of a weather pattern in seconds to predict the path of a deadly storm better.

The automation of intelligence with generative AI will transform every company in the world's $100 trillion worth of industries. Startups and established companies spanning every conceivable domain and industry are investing billions of dollars to advance and apply generative AI.

With AI supercomputers, we are manufacturing intelligence in the form of tokens or floating-point numbers. In the last industrial revolution, power generation plants produced electricity. In this new industrial revolution, NVIDIA AI supercomputers are the AI factories that produce intelligence. Generative AI has enabled us to produce something completely new.

Companies will operate a new type of facility, owned or rented, that produces AI to augment, accelerate, and amplify the productivity of their employees and business processes.

Nations have awakened to AI's impact. Countries are building sovereign AI infrastructures. Governments see AI factories as essential infrastructure to advance their society and local industries. Nations such as Canada, France, India, Singapore, Japan, and others seek to develop their sovereign AI capabilities as they have recognized their country's data is their most valuable natural resource.